YENEPOYA INSTITUTE OF ARTS, SCIENCE, COMMERCE AND MANAGEMENT A CONSTITUENT UNIT OF YENEPOYA (DEEMED TO BE UNIVERSITY) BALMATTA, MANGALORE

HIGH LEVEL DESIGN

ON

NATURAL LANGUAGE INTERFACE FOR EDTECH INFRASTRUCTURE

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1. Introduction

1.1 Scope of the Document

This document presents a high-level overview of the proposed EdTech query system. It explains how different components of the system interact and how user queries are processed to generate appropriate responses.

1.2 Intended Audience

· Project evaluators
· Development team
· Academic supervisors
· Individuals interested in system design

1.3 System Overview

The system is designed as a web-based application where users can enter queries in simple natural language. The application interprets the input, identifies user intent, and provides relevant EdTech-related information such as available courses, progress status, or platform support details.

2. System Design

2.1 Application Design

The system follows a layered architecture:
· Frontend Layer – Handles user interaction
· Backend Layer – Processes requests using Flask
· Processing Layer – Performs keyword-based NLP
· Data Layer – Stores and retrieves EdTech data

2.2 Process Flow

User enters a query →
Input is processed and cleaned →
Intent is identified →
Relevant data is retrieved →
Response is generated and displayed

2.3 Information Flow

· Input flows from user interface to backend
· Backend sends data to processing module
· Processing module identifies intent
· Data module retrieves required information
· Final response is returned to user

2.4 Components Design

The system consists of:
· User Interface Module – Accepts user queries
· Server Module (Flask) – Handles requests
· Intent Detection Module – Processes input
· Data Module – Stores EdTech information
· Response Module – Generates output
· Logging Module – Stores user interactions

2.5 Key Design Considerations

· Simple and user-friendly design
· Fast response time
· Modular architecture for scalability
· Logging for monitoring system usage

2.6 API Catalogue

Endpoint	Method	Purpose
/	GET	Displays homepage
/chat	POST	Processes user query

3. Data Design

3.1 Data Model

Data is stored in structured format (JSON/CSV), including:
· Course details
· Student progress
· Platform status

3.2 Data Access Mechanism

The system reads data using Python and processes it dynamically when queries are received.

3.3 Data Retention Policies

Data is stored locally. User queries are logged for analysis and debugging purposes.

3.4 Data Migration

Future versions can integrate databases such as MySQL or MongoDB.

4. Interfaces

· Web-based user interface
· Backend API for processing

5. State and Session Management

The system operates in a stateless manner, where each request is processed independently.

6. Caching

Caching is not implemented but can be added in future versions.

7. Non-Functional Requirements

7.1 Security Aspects

· Basic input validation
· Prevention of invalid inputs

7.2 Performance Aspects

· Fast response generation
· Lightweight processing logic

8. References

· Python Documentation
· Flask Documentation
· Basic NLP Concepts